BiomedGPT: A Generalist Vision-Language Foundation Model for Diverse Biomedical Tasks

Presenter: Jingyun Jia[1]

1 Department of Statistics, University of Wisconsin–Madison

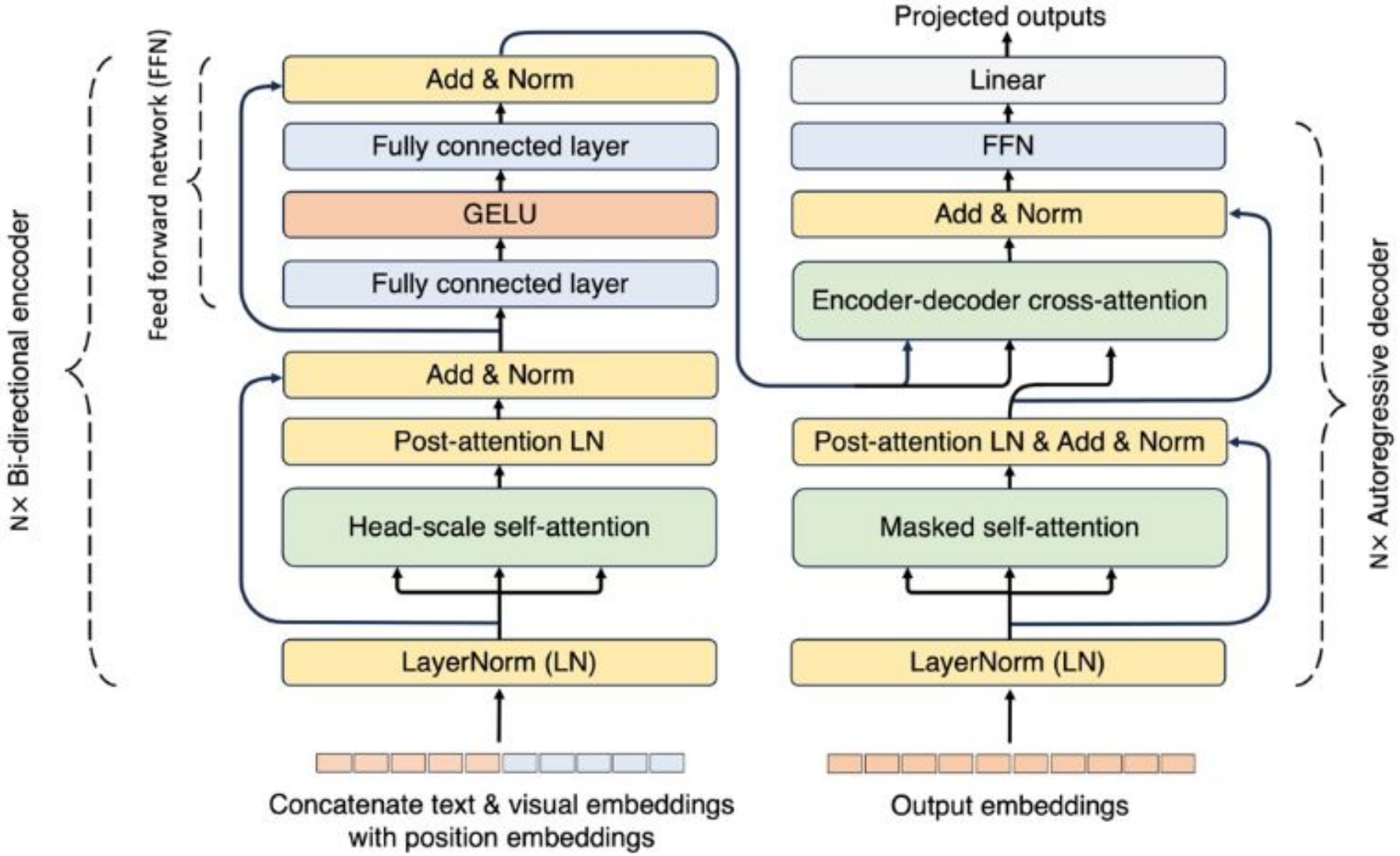
Background

- **Most biomedical AI models are specialist models designed for one task or one modality.**
- **Real biomedical applications often involve heterogeneous inputs and outputs: images, text, question answering, report generation, summarization, and classification.**
- **Maintaining separate task-specific models is inefficient and limits knowledge sharing.**

Key Challenges

- **Inputs can be image, text, or image + text.**
- **Outputs can be labels, captions, answers, summaries, reports, or detected objects.**

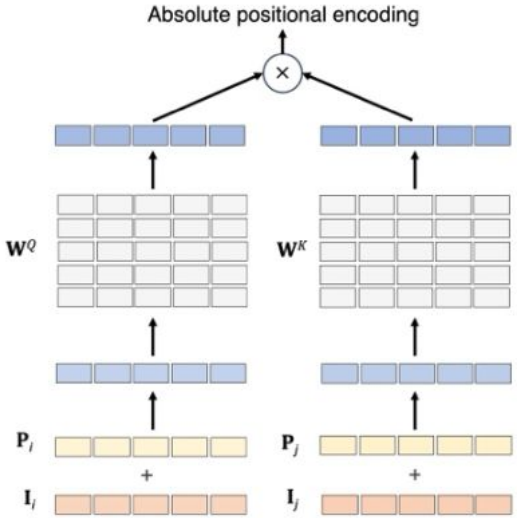
Overview of Architecture

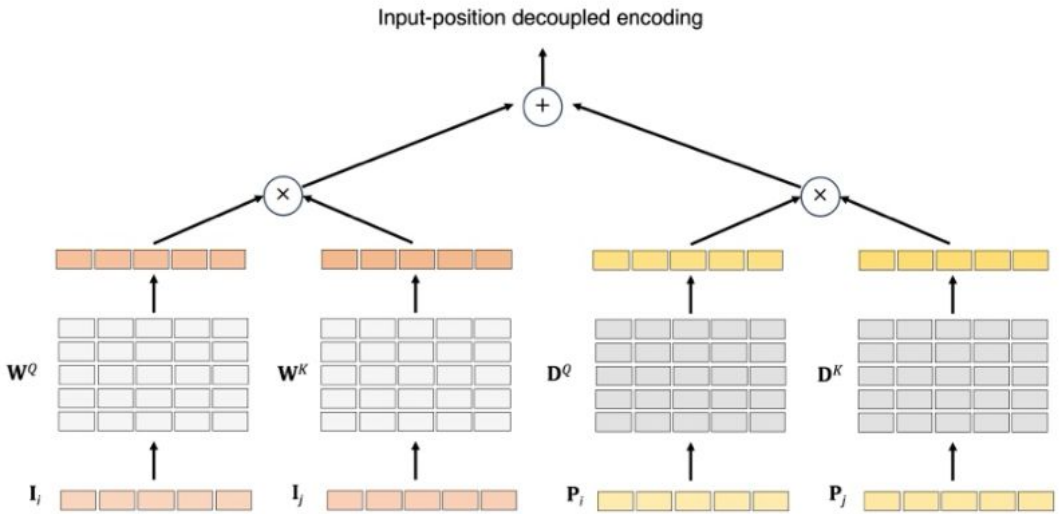


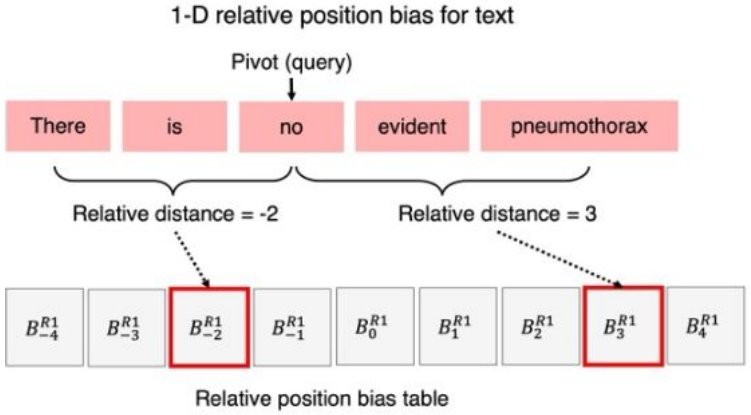
Key Components

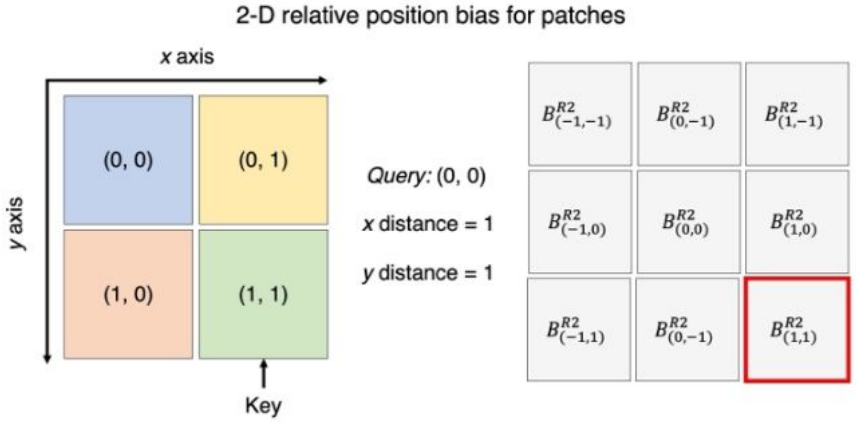
$$s_{ij} = \frac{(I_i W^Q)(I_j W^K)^\top + (P_i D^Q)(P_j D^K)^\top}{\sqrt{d}} + B^R_{ij}.$$

- semantic relationships
- absolute positional relationships
- relative positional bias.

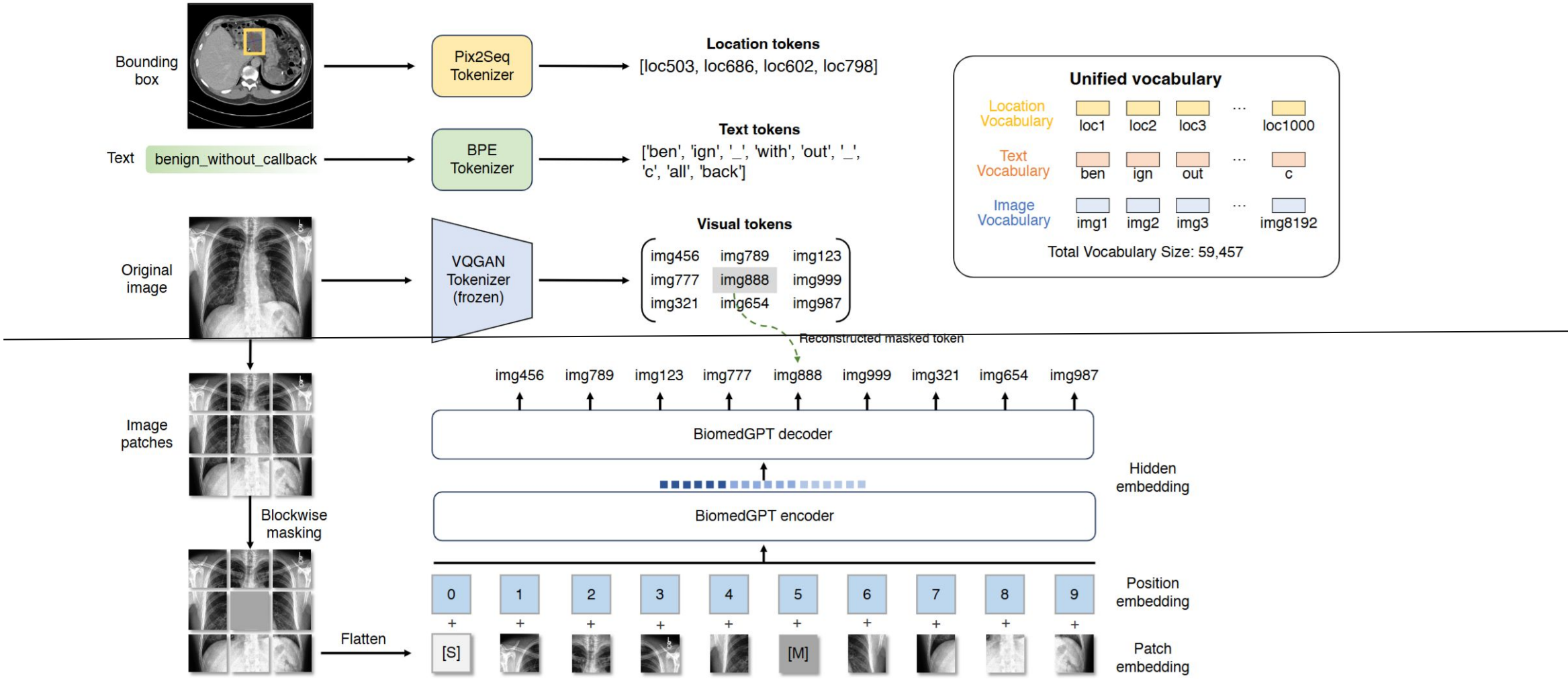






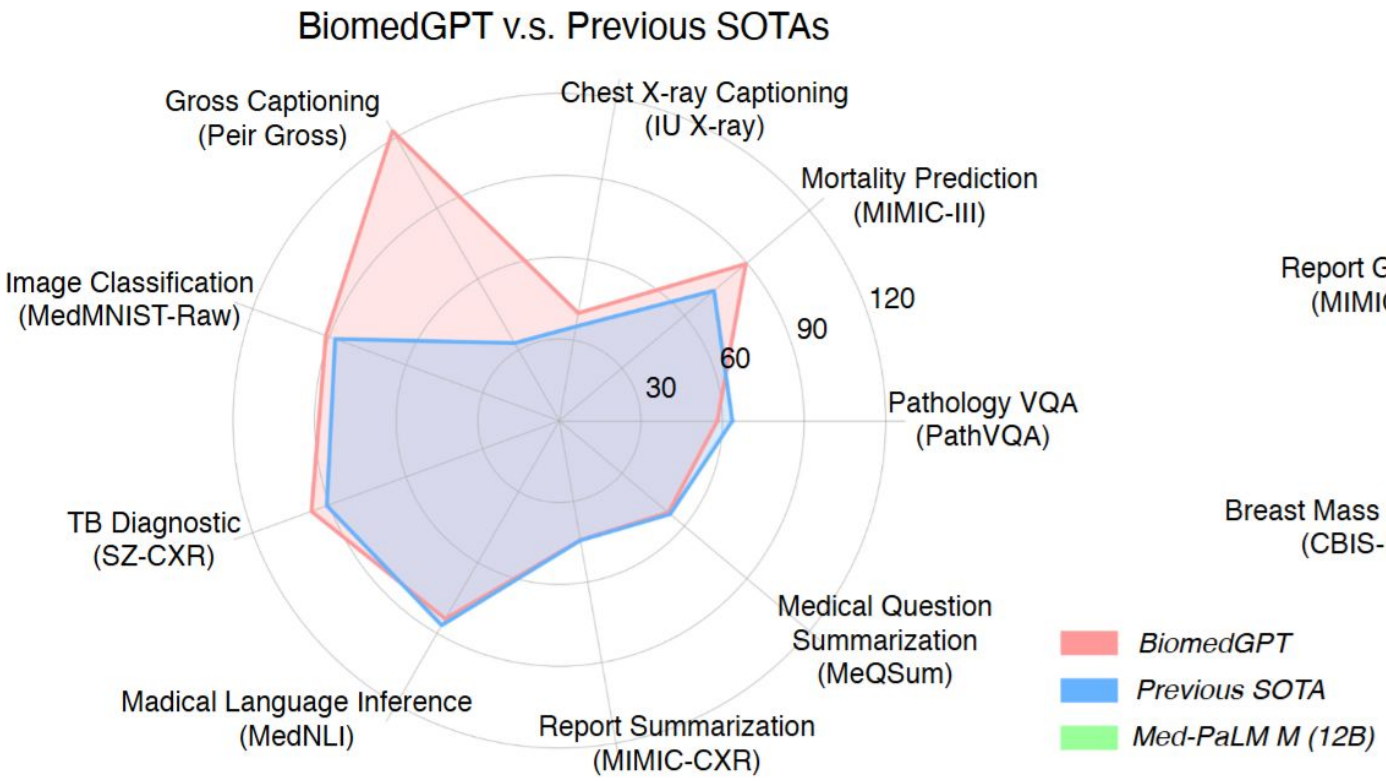


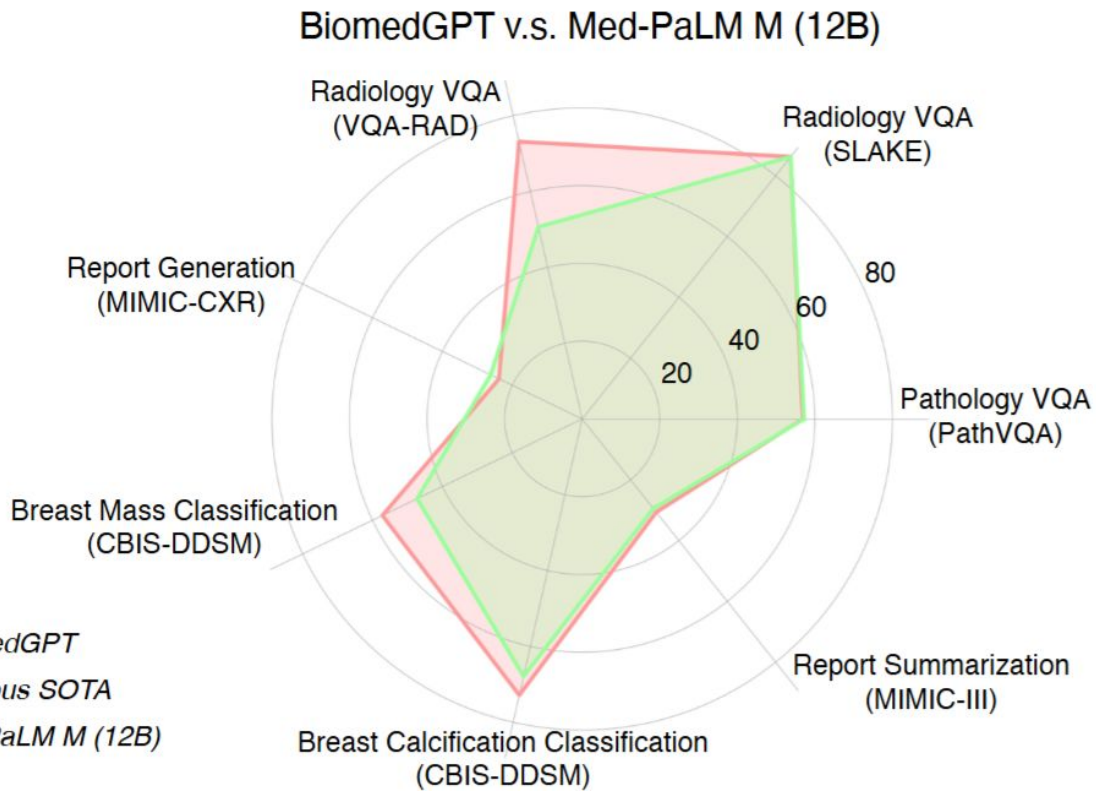
Key idea: Unified Token Space



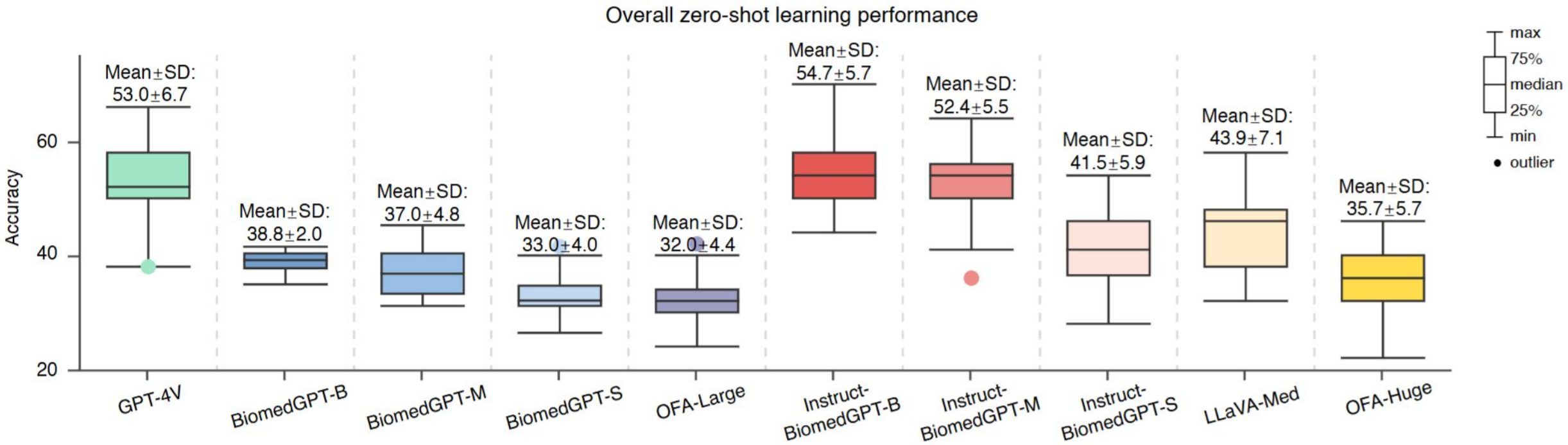
Benefit from Pre-training

182 M v.s. 12 B



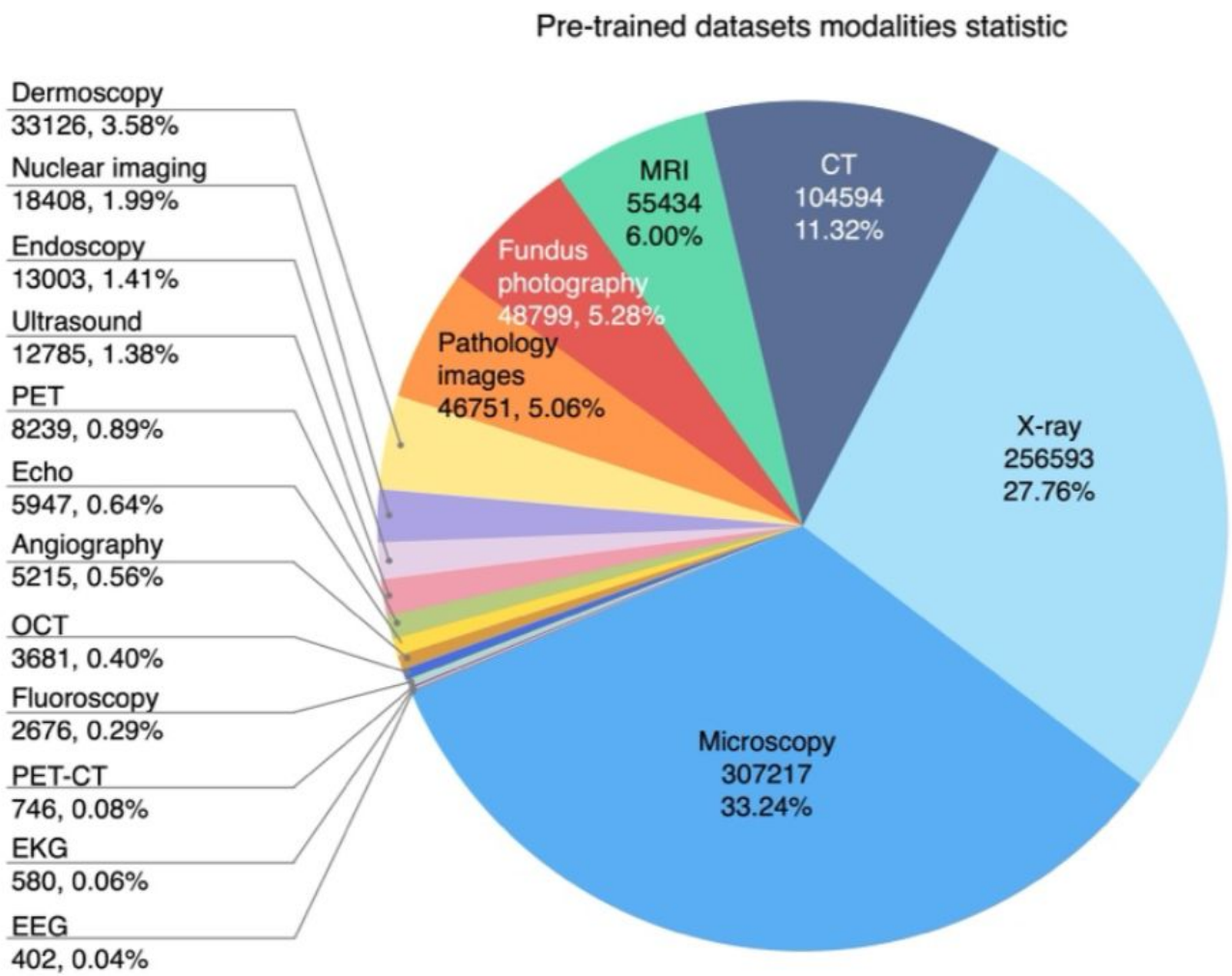


Zero-shot Inference Capability



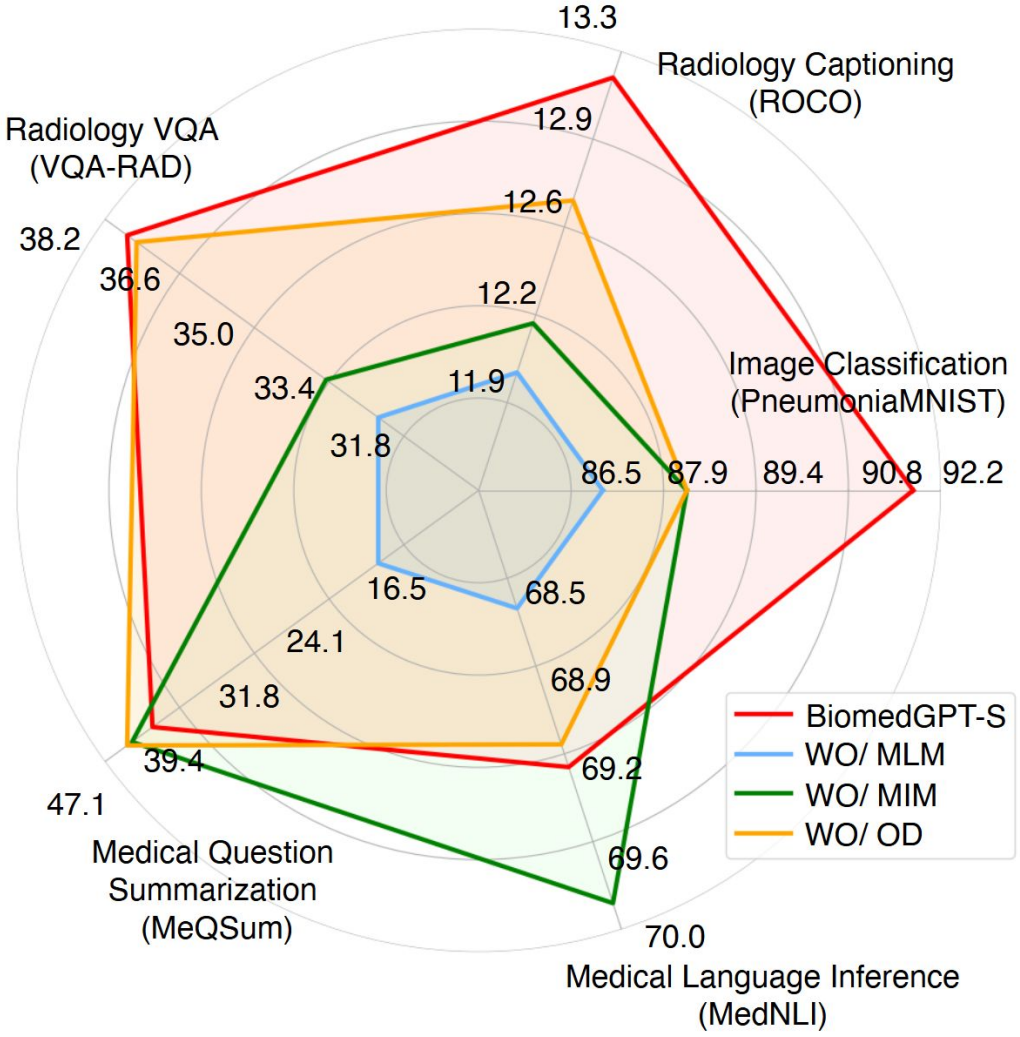
Discussion

More balanced data for pre-training



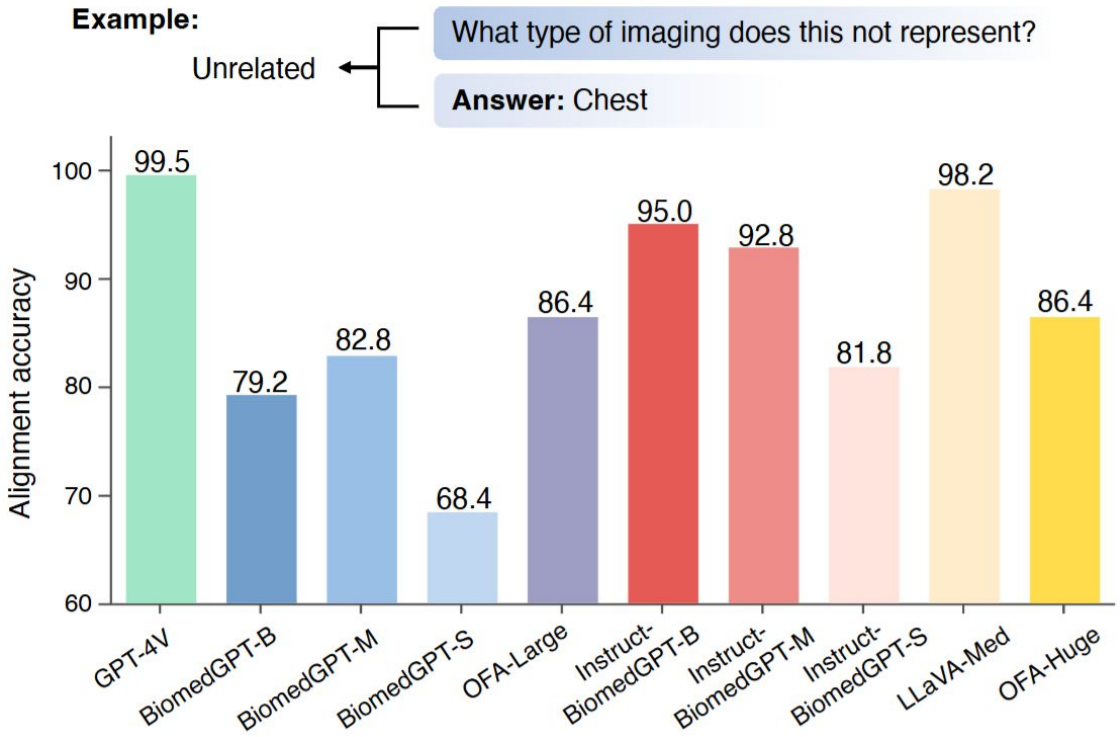
Discussion

Negative transfer due to heterogeneous modalities



Discussion

Current zero-shot models are not Reliable in Real World.



Average zero-shot accuracy across 7 question types

	GPT-4V	BiomedGPT-B	BiomedGPT-M	BiomedGPT-S	OFA-Large	Instruct-BiomedGPT-B	Instruct-BiomedGPT-M	Instruct-BiomedGPT-S	LLaVA-Med	OFA-Huge
Disease diagnosis	50.9	43.5	35.4	45.2	32.8	53.9	52.1	32.6	42.0	34.5
Imaging technical details	73.3	41.0	19.9	19.5	20.9	68.8	68.0	58.1	67.6	20.4
Lesion & abnormality detection	48.6	37.2	38.7	41.2	45.5	49.5	52.9	45.9	40.6	40.6
Modality recognition	77.9	68.7	59.6	42.7	43.4	77.1	69.5	55.5	69.4	55.0
Size assessment	46.6	39.7	59.4	37.8	42.9	44.6	65.3	39.5	68.6	44.6
Spatial relationships	47.7	14.4	21.8	9.5	23.6	44.0	31.8	27.6	35.4	28.2
Structural identification	52.0	41.3	28.8	32.4	30.7	43.1	35.2	37.0	41.0	40.0